VOLUME 250 JULY 2026 NUMBER 01

MUMMY SLEUTHS | PIT VIPERS | THE USA AT 250

NATIONAL GEOGRAPHIC

BEYOND EARTH



HOW WE'RE PREPARING TO LIVE AND WORK ON THE MOON AND MARS

JUL 21 2026
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NATIONAL GEOGRAPHIC EXPLORERS

These contributors have received funding from the National Geographic Society, which is committed to illuminating and protecting the wonder of our world. Learn more about the Society's support of Explorers at *natgeo.com/impact*.



Mackenzie Calle (at left) takes a selfie with fellow crewmember Shriya Musuku in Utah.

PHOTO: MACKENZIE CALLE

Why Mackenzie Calle Donned a Space Suit for a Photo Assignment

→ Brooklyn-based photographer **Mackenzie Calle's** latest project for the magazine ("This Isn't Outer Space. Yet." p. 108) took her to rural Utah, where she participated in a simulated space mission—one of 16 concurrent events staged around the world last fall. Over two weeks at the Mars Desert Research Station, Calle wore space suits, practiced emergency drills, and traveled in rovers across sandy expanses as part of a test run for a future mission to the red planet. Normally an observer, Calle turned the lens on herself, using a drone and camera timers to record her trailblazing adventures. "I felt really out of my element," Calle admits. But she soon embraced the thrill of being "part of the story for the first time."

Davide Monteleone
"The Secrets That Mummies Keep," p. 16

The photographer, based in Milan and Zurich, visited Austria, Italy, and Bolivia as he followed a team of scientists pioneering new ways of studying the long dead. In Bolivia he used ultraviolet and infrared light to reveal details of mummies that were otherwise undetectable. The experience, he says, offered an "almost intimate encounter with the past."

Chris Gunn
"Wonders of America," p. 46

For this issue, Gunn traveled from Washington, D.C., to photograph two places with quite different claims to fame: Crystal Cave in Put-in-Bay, Ohio, site of an enormous walk-in geode, and Missouri's Cathedral Basilica of St. Louis, home to the country's largest mosaic installation. "No amount of research prepared me for the scale and brilliance of the mosaics," says Gunn.



From March to October, Big Bend's Casa Grande Peak is illuminated at night by the Milky Way.

AMANDA FRIEDMAN

THE SNAKE THAT SAVES MOUNTAINS

High in the cloud forests of western Honduras, the stunning and elusive jade palm pit viper has become a symbol of a vanishing ecosystem—and now a critical part of a new conservation effort.

WORDS BY
LIZZIE WADE

PHOTOGRAPHS BY
JACK DEVLIN

The emerald-colored scales of the jade palm pit viper allow it to blend into the damp foliage of the woodlands where it lives. In recent years, much of its Central American habitat has been destroyed to make way for coffee plantations and pasture for cattle.

VANTAGE POINT

36° 44' 13" N, 2° 43' 3" W

Glimpsing the FUTURE *of Fresh* FOOD

Along Spain's southern coast, a peninsula teeming with greenhouses offers a revealing look at how Europeans are getting fed.

Words by SAM KEAN



Satellite map: Matthew W. Chwastyk
SOURCES: PLANET LABS; NASA

THIS ISN'T OUTER SPACE.

YET.

Before humans can venture to other worlds—or set up colonies on the moon—there's a lot to be rehearsed. Inside the increasingly realistic and dramatically complex experience of simulated space travel.

WORDS BY
ERIN BERGER

PHOTOGRAPHS BY
MACKENZIE CALLE AND
CASSANDRA KLOS

In preparation for the World's Biggest Analog, an audacious experiment that placed earthbound space advocates in the roles of tomorrow's astronauts, crewmembers of the Mars Desert Research Station (MDRS) run a pre-mission exercise in the desert outside Hanksville, Utah.

For two weeks, 76 people lived in 16 habitats designed to simulate prolonged space expeditions on, or travel to, the moon and Mars. At MDRS, crew scientist Shriya Musuku (at left) prepares to exit the base while crew engineer Ricardo J. Gonzalez and commander Rym Y. Chaïd take a space walk (at right).

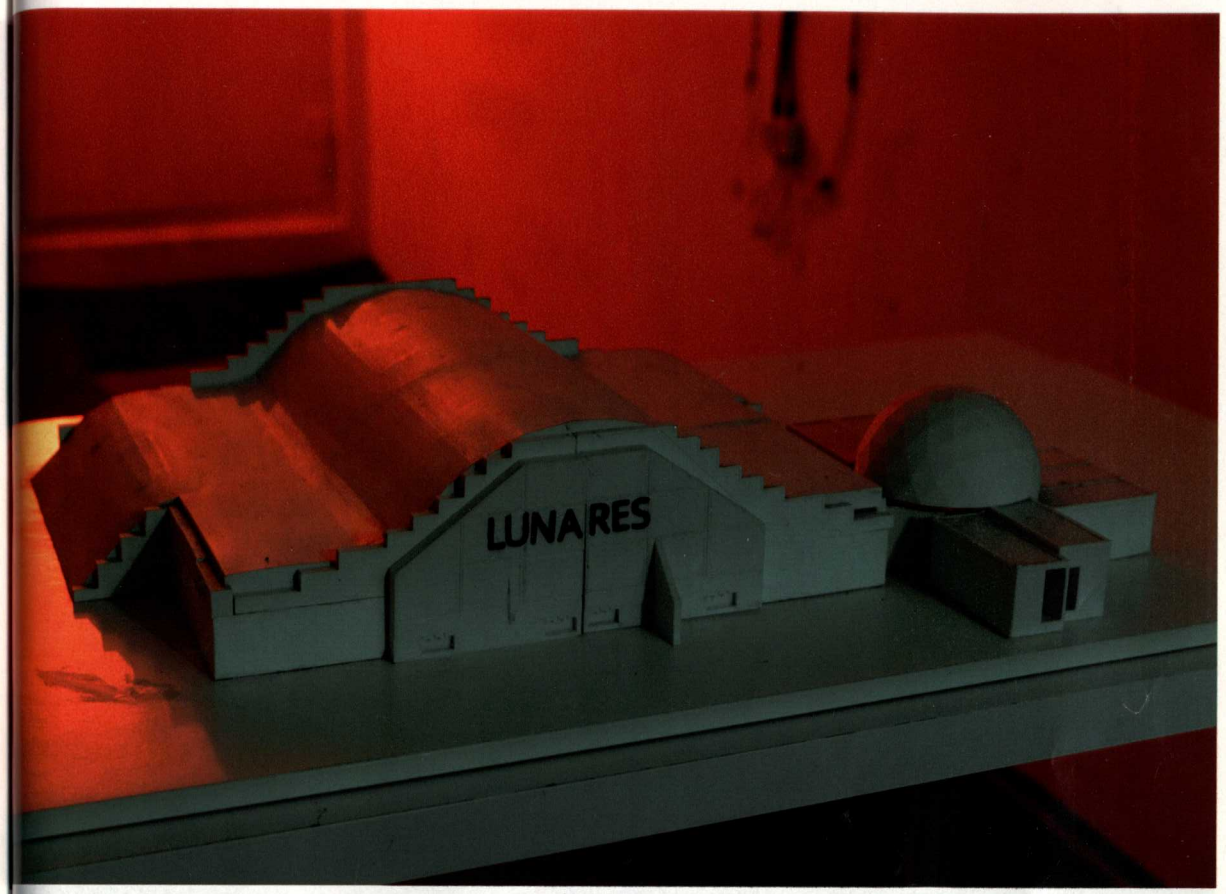
No major national space programs were involved in running the World's Biggest Analog, making it the most complex independent simulation of its kind. The crews had a sense of autonomy as they chose some of their research exercises across multiple sites. In Utah, MDRS participants including Chaïd, shown here, conducted soil sampling.

In space, death is always an inch away. An unsecured airlock, a crack in a helmet, a single loose screw. In the face of an emergency, there is no time for a mistake and even less time to panic. It's why astronauts are trained to be creatures of regimen and protocol—even when those astronauts are participating in a simulated space mission at a massive airport hangar in Piła, Poland.

One day last fall, three members of a six-person team sealed their space suits and exited the hatch of their cramped artificial moon base, a bunkerlike habitat adjoining the 3,000-square-foot, completely fabricated lunar landscape at LunAres Research Station. The Polish scientific facility was sealed shut and pitch-black, designed for this crew specifically to replicate the experience of living at the south pole of the moon.

This particular test site is just one of many places around the globe now dedicated to modeling what astronauts may encounter within extraterrestrial environments, a format that researchers often refer to as a "space analog." In this case, the crew was heading out on an extravehicular activity, or EVA, to take samples of the terrain—a mix of rocks and gravel to replicate mineral-rich regolith. They were also taking part in an experiment of even grander scale.

The LunAres crew had joined with participants from a total of 16 different space analogs running trials simultaneously at scientific sites worldwide as part of an ambitious, international initiative called the World's Biggest Analog. The goal: To test not only many kinds





At LunAres Research Station in Piła, Poland, one crew of astronauts took part in a simulation replicating the total darkness of a crater at the moon's southern pole. To achieve that effect, their artificial moon environment was in a sealed 3,000-square-foot aircraft hangar. A 3D-printed model of LunAres (above) sits inside the station's airlock (below).



Crew science officer Nicole Bolado worked on growing microgreens during her time at LunAres. Without access to direct sunlight, the team tested new ways to extend their food supply beyond the rations they'd packed for the experiment.

of future space missions but also how such operations might stay connected while happening in multiple places across the solar system—specifically, humanity's next two destinations, the moon and Mars.

Earlier this year, NASA's Artemis program sent a team of astronauts around the moon and farther than any crewed mission before, kicking off a new era of space exploration. In 2028 Artemis is expected to lead the first moonwalking expedition in over 50 years. Crewed missions to Mars may begin within the next decade, whether by space agencies or private space companies, which could redefine what the future of extraterrestrial habitation, scientific research, and mining off-planet resources looks like. But all of it depends on making sure that a new vanguard of astronauts will be prepared for the challenges of extended space navigation.

"A miscommunication or a failure to coordinate effectively is going to have not just potential to harm the crew or to have implications for research," says Marissa Shuffler, an organizational psychologist at Clemson University who has worked with NASA on its own earthbound space trials. "A failure could set back programs by quite a bit."

That's where the World's Biggest Analog comes in. The idea originated several years ago with a U.K.-based physics researcher named Jas Purewal, who knew that while many national space programs run their own analogs to prepare future astronauts for some rigors and challenges of space travel, there's simply no way to test everything an astronaut might encounter. (In fact, NASA refers to such experiments exclusively as "analogs," not simulations, because they don't re-create all the variables of space.) At the same time, she noticed that a growing universe of publicly accessible space analogs have been established, creating a shared place for space agencies, private companies, and academic researchers to safely test different scenarios ahead of an actual launch. Through closely coordinated testing, Purewal reasoned, her group might begin to "actually address some of these research gaps."

To do that, Purewal and a small team of volunteers solicited public donations and held an open call for anyone interested in becoming an Earth-based astronaut. If chosen, many analog astronauts agreed to pay a couple thousand dollars to help fund facility costs. The response was overwhelming: Around 500 people applied and underwent a thorough prequalification process aimed in part at gaining insights into how people think and perform under pressure, another skill that's inevitably tested in space.

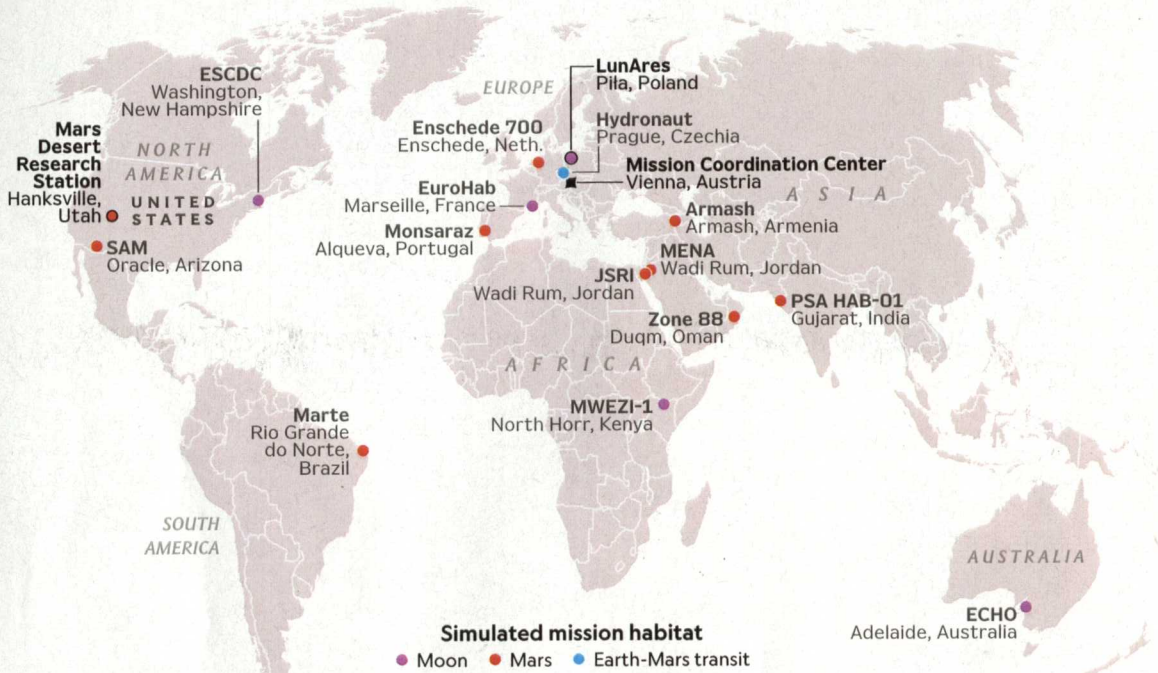
Case in point: An hour into the LunAres team's moon walk, they received an alert over their helmet radios. One of the crewmembers monitoring their progress had registered that someone's vitals were tanking. Medical officer Domina Stamas and science officer Nicole Bolado discovered that their third crewmember—a space suit–clad test dummy named Emma—was lying prone on the surface.

The nature of her injury was unclear, so they needed to work quickly to drag her back inside the airlock for evaluation. Stamas and Bolado knew what to do; they'd drilled for this scenario. Bolado reached into her backpack and retrieved a collapsible, lightweight gurney. Together, the astronauts tried to assemble the stretcher

Inside the MDRS base, Gonzalez takes time to stretch as part of a regular exercise regimen. Finding ways to maintain mental and physical health while coping with isolation and confinement was another focus of the simulated missions.

MAKING SPACE

To prepare for the future of space travel, researchers launched the World's Biggest Analog: a global experiment with 16 simulations, exploring challenges we may soon face on the moon and Mars.



by lacing together a series of cords. As precious seconds ticked away, though, they seemed to be struggling.

"HabCom, this is EVA 1—we have misidentified some of the loops," Stamas radioed back to base. "Can you talk us through how to correctly loop this gurney?"

THE MODERN ERA of simulated space missions started with a few small steps in the Southern California desert. In 1997, not long before the landing of the first ever successful Mars rover, Sojourner, NASA sent a crew of four, including members of the Johnson Space Center Advanced EVA team, to Death Valley to prepare for unplanned, complex missions that might one day take its astronauts across the solar system. The agency had concluded that the arid, sandy environment was a prime location to conduct speculative fieldwork.

Over the past three decades, proactive space planning has experienced something of a Cambrian explosion. Today NASA's Human Research Program is involved with about 10 analog test facilities that intersect with how we might better cope with concerns like low-gravity environments, motion sickness, and space radiation exposure. Across the globe, space agencies and private companies have also run simulations in extreme environments that range from the ocean to underground lava tubes and Antarctica. These simulations serve another essential purpose: spotting new ways to avoid potential disasters caused by tiny, cascading human errors.

In 2013, for example, Italian astronaut Luca Parmitano experienced a water leak in his helmet during a space walk from the International Space Station, forcing him to feel his way back to the airlock with blurred vision. NASA's incident report found, among other miscommunications, that ground controllers brushed off Parmitano's mention of a sensor failure early in the space walk because it had happened before without consequence.

Ignoring a recurring potential concern is known as normalization of deviance, a thought process that, over the decades, has led to the space agency's most significant tragedies. The 1986 *Challenger* explosion occurred after seals failed in low launch temperatures, and the 2003 *Columbia* shuttle disintegrated during reentry after being struck by foam insulation at liftoff. In both cases, the issues had been flagged and deemed acceptable risks.

The problem with space exploration is that there are always new issues to troubleshoot. "There's just not enough time or missions actually occurring in space ... to do every study," says Brandon Vessey, a scientist in NASA's Human Research Program who provides oversight on projects involving analogs and spaceflight.

Independent analogs can help solve that. In one recent study, for instance, researchers used mobile eye-tracking technology to study changes in visual performance and related cognitive demands for astronauts dealing with prolonged time in an enclosed habitat. In another, aimed at growing our extraterrestrial workforce, researchers measured the activities of an analog crew that included one person with a below-the-knee prosthetic to determine how they performed the active duties required to operate an analog mission.

When it comes to potentially catastrophic issues being overlooked, NASA's own simulations have helped explain how faulty thinking can happen in stressful environments during prolonged space travel. In an agency-funded study published in 2025, Northwestern University researchers discovered that while crews get better over time at familiar tasks like repairs, they actually become worse at creative thinking and navigating unexpected problems.

Such findings have made the agency even more watchful of what's happening in private simulations as they seek to understand how, among other things, group chemistry might lead to more success during trying times. "I have a lot of interest in the isolation and confinement analogs, just because of the complexity of getting people into a mission mindset that's sufficient to believe that the behaviors you're seeing are similar to what you would've seen on a real mission," Vessey says.

While there are some obvious drawbacks to earthbound research—like gravity, for instance—and still no widely accepted formal set of guidelines to govern all test scenarios, NASA remains supportive. "From our perspective," says Vessey, "the more folks out there contributing to prepping for space exploration, the better."

BACK AT LUNARES, as Stamas struggled with the gurney ropes, the communications team inside the base carefully walked her through the assembly process, explaining how to loop and secure a carabiner to ensure the gurney's stability. Even though the crew

The nonprofit National Geographic Society, committed to illuminating and protecting the wonder of our world, funded the work of photographer and Explorer Mackenzie Calle featured in this story.

MAP: CHRISTINE FELLENZ, NGM STAFF. SOURCE: AUSTRIAN SPACE FORUM

Many habitats used in the World's Biggest Analog are accessible for use year-round, including by other researchers seeking places that mimic off-Earth environments. Here, National Geographic Explorer Mackenzie Calle, Chaïd, and Gonzalez embark into the Martian desert of southeastern Utah.

During an expedition at MDRS, Musuku (at right) drives one of the base's four rovers with Calle riding shotgun. These missions were often focused on exploring the terrain, documenting the landscape, and testing the boundaries of communication signals.

recognized that they were trying to pull an imaginary person back to a makeshift airlock attached to a decommissioned Cold War–era hangar that had been filled with sand and rocks, the stress of the situation still felt real.

Everyone was also beginning to feel the effects of natural light deprivation. Cassandra Klos, a documentary photographer and crewmember, said that as the mission wore on, some things they'd done only hours earlier felt like days before, while events from the week prior seemed like they might have just happened. Still, by working together, the crew sped up the problem-solving process enough that it might have saved a real person's life.

Understanding why the initial confusion occurred is the kind of opportunity that Purewal, who has participated in emergency drills during other analogs, hoped to identify. "You find sort of the holes," she says about seeing what types of problems may arise.

For Emily Apollonio, the founder of Washington, D.C.-based Interstellar Performance Labs, which spearheaded the recruitment of many participants for the World's Biggest Analog, such moments may ultimately help reinforce the question that she finds essential for the next wave of outer space astronauts, regardless of their individual specialties. "Is this person coachable?" is one of her biggest criteria. In order to narrow down the initial pool, her team reviewed written materials from the roughly 500 applicants, selecting 200 candidates to participate in a series of group interviews; from there, about 70 people participated in three months of virtual and in-person team building exercises. To determine their final selections, Apollonio's team looked specifically for crew members who could demonstrate that they were adaptable and capable of collaborative problem-solving.

"It's really a game of trying to figure out which groups and individuals are going to work best with each other," Apollonio says, adding that the primary thing her team looked for was people's "willingness to learn" and get to know others.

Beyond the teamwork happening at each habitat, the World's Biggest Analog also tested how an interplanetary mission control might work one day. The entire off-Earth network of bases coordinated with one central command of about 40 operators located at the Austrian Space Forum, a nonprofit public



The hangar at LunAres is filled with dirt and rocks to replicate the moon's surface. During the World's Biggest Analog, a crew of six analog astronauts, including commander Shaziana Kaderali (at left) and crew engineer Makiah Eustice, conducted research with the assistance of a small rover named Leo.

All the crewmembers on LunAres, including Eustice (left), were required to wear space suits while outside their bases. Back inside, a view of Earth through a porthole window (right) was approximated by hanging an homage to the famous 1968 "Earthrise" photograph.

science group headquartered in Vienna, which then relayed critical information to smaller mission control teams at each of the 16 sites.

The team at the Vienna hub checked in on each habitat's scientific experiments, shared daily weather reports, collected mission logs, and connected similar analogs with each other for morale-boosting video chats. They also documented the day's tasks in exhaustive detail, just as a real universal mission control would need to stay running for 24 hours a day to keep people throughout the solar system connected.

"One of the most important takeaways... was the immense depth of trust that had to be established to work together across [six] continents," says Gernot Groemer, the Austrian Space Forum director who led the interplanetary coordination. "Space exploration is a team sport, international, intercultural, interdisciplinary, sometimes even intergenerational."

Altogether, humanity has spent just a collective 12 days on the surface of the moon. And down to the hour, all six Apollo lunar landing missions were intensely regimented, with near-constant communication with earthbound staff. For Groemer and others at the World's Biggest Analog central mission support, the increased autonomy of future deep space crews presented an opportunity to allow participants to explore what will likely be a critical aspect of space exploration: the freedom to chart their own course after touchdown.

THE ASTRONAUTS of tomorrow are going to have a lot more to do once they arrive at their destination, and especially with deep space communication delays, they may have less live support from mission control than ever before. Ambitions might involve living on-planet for weeks at a time, building habitations, growing crops, and exploring very far afield. These have never been the main responsibilities of astronauts. So, as the LunAres team conducted its work from a dark moon crater and 15 other simulation crews operated from additional interstellar outposts and spacecraft, defining both autonomy and crew protocol became a key part of the collective mission.

This was particularly true at the Mars Desert Research Station (MDRS) in Utah, where a crew of four analog astronauts, including documentary photographer and National Geographic Explorer Mackenzie Calle, carved their own path across the red planet. Some days, two crewmembers would exit their habitat, climb onto a rover, and ramble for up to two or three hours at a time over smooth, sun-drenched red hills near Capitol Reef National Park.

They tested and chose their preferred space suits, gathered geological samples for lab analysis, and practiced emergency drills in the field. For the MDRS team, this was all strategic. Astronauts planning for extended stays on the moon or Mars will have to specialize in more roles like emergency medicine or engineering life-sustaining resources.

"As the number of jobs in space increases, it will be a community," Calle says. "You are going to have personalities, you are going to have differences, and you are going to be close. Those are the only people you're going to have."

All of which makes team chemistry matter even more. That was put to the test one afternoon when Calle and crew scientist Shriya Musuku were in the rover for what had been planned as a standard trip documenting an unexplored region known as the Sea of Shells. Suddenly, they lost contact with their base. Unlike the LunAres medical drill, this was not a simulated emergency. Although they'd been traveling for less than 30 minutes, the sand had deepened, and, with no GPS or communications link, the way back to base wasn't readily apparent.

"We kind of just realized that we're out here alone, and we have to figure it out," says Musuku. "That probably is one of the worst-case situations if you're out on Mars."

Calle and Musuku decided they must agree completely on each successive step from there on out. They agreed to stop the rover and reset the comms equipment. No luck. With the way back unclear, their next best bet was to find high ground and try to spot North Ridge, a striated domelike section of the desert terrain close to the base. "If you see that, you know where home is," Musuku says.

The pair identified an elevated ledge and slowly navigated the rover up it. At the top, just a few feet from a roughly seven-foot drop-off, they spotted North Ridge. Around the same time, the communications team buzzed over their radio. It had been roughly 15 minutes since their last check-in, and they were "pretty panicked," according to Musuku. After discussing the plan and their route with the comms team, Musuku made what she estimated was a 16-point turn on the narrow ledge, with Calle acting as her spotter, and followed their tire marks back to where they'd lost the main road.

Once safely back at base, the expedition crew and mission control team debriefed. One immediate lesson was that they needed to establish a more concrete protocol for how to dispatch a search party. Another might be that Apollonio's effort to prioritize teamwork had once again proved out.

ONE DAY IN THE not too distant future, humanity may build a real moon base that serves as an off-planet analog for Mars, allowing us to understand the granular details of deep space living while being only a few days' journey back to Earth should something go wrong. But whether the future simulations—on Earth or beyond—are run by NASA-backed specialists or private groups, we will undoubtedly carry our human quirks with us wherever we venture.

For their final night in outer space, the MDRS crew cooked one of their last dehydrated meals together. Meanwhile, at LunAres, the team found ways to stay upbeat, hosting karaoke nights and concocting a set of long-running jokes around Lunar Barbie, based on the popularity of the recent Barbie movie. On their last day, they stood in the fuchsia glow of the biolab's plant lights and counted down from 10 before opening the door and letting out whoops as they emerged into daylight.

The small details might sound trivial, but each communal action kept spirits high as the crew embraced their new reality. One evening, back at the Mars base, Calle found herself looking out at the night sky, imagining a time when someone like her could be stationed among the stars. "It almost felt like... one of those little dots on the horizon I was looking at was Earth," she says. "There could be any one of a million little pale blue dots I was looking at, but one of them was our whole lives." Simulation or not, the future felt inspiring. ◻

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